Radical Ideas

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This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were prepared in *ChemDoodle*.

"It's not the answer – it's the question that drives us." With that faux-thoughtful phrase from *The Matrix* (1999) in mind, we will explore these following questions about radical reactions. Some of these questions review essential skills from previous courses, other are meant to "free your mind"[1]

[1] See the movie for more of these deep-yet-shallow philosophical one-liners as they are spoken randomly between slo-mo kung-fu gunfights.

Before Class: A Review of Kinetics

Before we dive into radical clocks as a method in physical organic chemistry let us answer the following questions. Reach back to your first year chemistry and your two physical chemistry courses.[2] You got this. Attempt the following questions before the class meeting.

[2] You will need to consult your notes from previous courses. Ch. 7.4 & 7.5 of the textbook may help.

1. In the reaction scheme at the right in figure 1, I observe two possible products from the same reactants. Derive a relationship between the observed product ratio of $[C]/[D]$ and the rate constants k_1 and k_2.

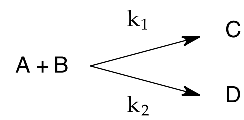

Figure 1: Scheme for question 1

2. Obtain a rate law for the formation of product in the reaction scheme presented at the right in figure 2 by using the steady-state assumption.
   (a) If the first step of the reaction is rapid and reversible, the rate expression can be simplified. What is the rate expression in this case?
   (b) If $k_2 \gg k_{-1}$, what does the rate expression simplify to?
   (c) Comment on these two cases above. Is the steady state approach a general case from which all reactions with an intermediate can be explained, or not?

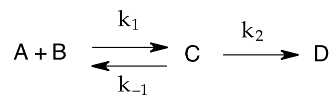

Figure 2: Scheme for question 2

3. Radial chain reactions can have interesting rate laws. In high-yielding reactions the initiation and termination reactions are unimportant in analyzing product ratios, but the termination reaction can have a consequence in the reaction kinetics. Presented in table 1 below are three radial chain reaction schemes for the chlorination of methane that differ only in the termination step. Determine the reaction order in $[Cl_2]$ and in $[CH_4]$ for each scheme.

$$CH_4 + Cl_2 \longrightarrow HCl + CH_3Cl$$

Figure 3: Scheme for question 3

Table 1: Three possible reaction schemes for question #3

Scheme #1	Scheme #2	Scheme #3
$Cl_2 \xrightarrow{k_1} Cl\cdot + Cl\cdot$	$Cl_2 \xrightarrow{k_1} Cl\cdot + Cl\cdot$	$Cl_2 \xrightarrow{k_1} Cl\cdot + Cl\cdot$
$CH_4 + Cl\cdot \xrightarrow{k_2} HCl + \cdot CH_3$	$CH_4 + Cl\cdot \xrightarrow{k_2} HCl + \cdot CH_3$	$CH_4 + Cl\cdot \xrightarrow{k_2} HCl + \cdot CH_3$
$\cdot CH_3 + Cl_2 \xrightarrow{k_3} CH_3Cl + Cl\cdot$	$\cdot CH_3 + Cl_2 \xrightarrow{k_3} CH_3Cl + Cl\cdot$	$\cdot CH_3 + Cl_2 \xrightarrow{k_3} CH_3Cl + Cl\cdot$
$Cl\cdot + Cl\cdot \xrightarrow{k_4} Cl_2$	$\cdot CH_3 + \cdot CH_3 \xrightarrow{k_4} C_2H_6$	$\cdot CH_3 + Cl\cdot \xrightarrow{k_4} CH_3Cl$

The following page is based upon "Evidence for free-radical reductive dehalogenation in reaction of zinc and acid with 1-perfluoroalkyl-2-iodoalkanes and with 1-perfluoroalkyl-2-iodoalkenes", N.O. Brace, J.E. Van Elswyk, *J. Org. Chem.*, **1976**, *41*, 766–771. https://doi.org/10.1021/jo00867a004.
For a review of radical clocks see "Free-radical clocks", D.Griller, K.U. Ingold, *Acc. Chem. Res.*, **1980**, *13*, 317–323. https://doi.org/10.1021/ar50153a004

Class Meeting: Tick Tock.

Chapter 8 covers many methods of investigating reactions in detail. You have been introduced with the concept of radical clocks. Let us take a moment to explore this subject some more.

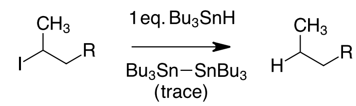

Figure 4: Scheme for question 4

4. We can reduce alkyl halides to alkanes easily using a radical dehalogenation reaction involving tributyltin hydride. A small amount of bis(tributyltin) is required for this reaction to proceed. See figure 4.
   (a) Propose a reaction mechanism for this reaction.
   (b) Propose a rate law for this reaction.

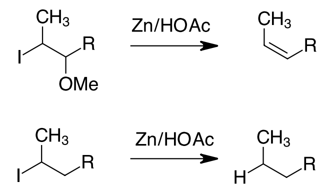

Figure 5: Scheme for question 5

5. The same result has been accomplished using a dissolving-metal reaction. This is believed to proceed via a Grignard intermediate. See figure 5.
   (a) Given the evidence from the two reactions above, propose a mechanism for this reaction.
   (b) Show how this mechanism is consistent with both reactions.
   (c) Propose a rate law for this reaction.

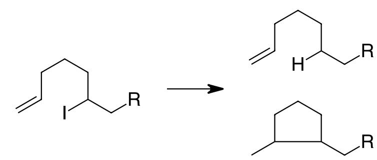

Figure 6: A radical clock experiment for problems 6 and 8: the hexaenyl clock.

Table 2: Product ratios for hexaenyl clock

Reaction	substitution	closure
HSnBu$_3$	33%	67%
Zn/HOAc	23%	77%

6. Some researchers believe that a transient radical intermediate exists in the pathway toward the Grignard. Consider the dehalogenation reactions in figures 6 and 7. The reactants were dehalogenated using both the tributyltin hydride reaction and the zinc Grignard conditions. Results are tabulated below each scheme. See figures 6 & 7 and tables 2 & 3.
   (a) Propose a mechanism for each reaction condition. Explain how the evidence is consistent with your proposal.
   (b) Propose an equation for the observed product ratio. Explain how this could be used to explore rates of reactions that occur after the RDS.

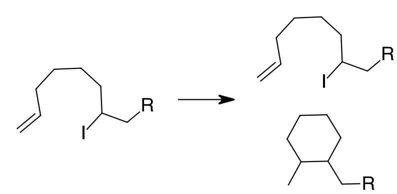

Figure 7: A different radical clock experiment for problems 6 and 8: the heptaenyl clock

Table 3: Product ratios for heptaenyl clock

Reaction	substitution	closure
HSnBu$_3$	98%	2%
Zn/HOAc	84%	16%

7. This experiment is an example of a radical clock. Clocks tell time. Radical clocks can tell you the lifetime of a radical intermediate. Tell me how a radical clock can be used to measure the lifetime of a radical.

8. Using table 8.7 of your textbook, calculate the relative rate of hydrogen abstraction from the tin hydride reagent by the radical intermediate in that reaction compared to the cyclization reaction of the hexaeneyl radical clock (See figure 6 and table 2.) Then use this result to calculate the rate constant for cyclization of the heptaenyl radical clock. (See figure 7 and table 3.) Congratulations, you have just added a clock to the *horlogerie* of radical clocks. (You should use that word if you own more than one wristwatch – impress your friends).

9. Using table 8.7 of your textbook, predict the results of the reactions in figure 8 (products and product ratios, assuming the radical substitution reaction rate does not change significantly.)

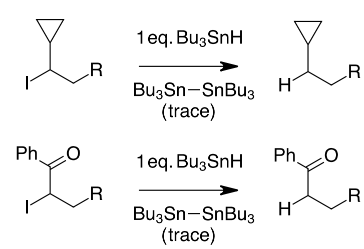

Figure 8: Reaction schemes for problem 9

Textbook Questions

As you complete your reading in chapter 8, try these textbook problems: Ch. 8: 21, 22, & 27

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will…

- have considered the many other possible experiments for examining reaction mechanisms outlined in the textbook.[3]
- have gained an understanding of how a competing reaction of known rate can be used to ballpark another reaction rate.
- be able to design an experiment to estimate the rate of a radical reaction by incorporating a radical clock into the reactant.
- be able to interpret the results of a radical clock experiment.
- have learned a new french word.

[3] Read ch. 8.8